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Optimization Algorithms for Retrieving the Bottomside Ionospheric *B*0 and *B*1 Parameters

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Key Points:

- A framework combining robust nonlinear least squares (NLS) and gradient optimization accurately retrieves bottomside ionospheric parameters
- The NLS-initialized limited-memory BFGS method exhibits the highest accuracy and computational stability among tested algorithms
- Evaluation of the ABT-2009 model reveals its systematic biases at high and low latitudes and high dispersion during low solar activity

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Abstract The bottomside morphological parameters of the ionospheric F2 layer, the thickness parameter *B*0 and the shape parameter *B*1, are crucial for describing electron density profiles and monitoring space weather. However, existing climatological models and indirect ionosonde approaches have limitations in real-time applicability and direct analytical fitting, struggling to meet the high-precision, continuous representation needs of three-dimensional ionospheric modeling. We propose a direct joint inversion framework based on ionosonde electron density data, combining robust nonlinear least-squares (NLS) initialization with modern gradient optimization algorithms, including gradient descent, Adam, AdaGrad, and limited-memory BFGS (L-BFGS). The convergence performance of these algorithms was systematically compared using global ionosonde observations from 2014 (high solar activity) and 2020 (low solar activity). Results demonstrated that the NLS-initialized L-BFGS approach performs best overall in inversion accuracy, convergence stability, and computational efficiency across various conditions. These optimized parameters were then used to quantitatively evaluate the ABT-2009 empirical model, revealing pronounced spatiotemporal variability in its performance. During high solar activity, the model exhibits systematic biases at low and high latitudes. Conversely, during low solar activity, prediction dispersion increases in summer mid-latitudes. Persistent error concentration in low-latitude and polar regions highlights the deficiencies of empirical models in capturing regional variability and dynamic physical processes. This high-precision framework accurately retrieves ionospheric parameters, effectively diagnosing existing model deficiencies and laying a solid data foundation for future three-dimensional ionospheric modeling.

1. Introduction

The ionospheric F2 layer is the region of peak electron density in the Earth's upper atmosphere, and its morphological variations have a decisive impact on many radio wave-dependent applications, including high-frequency (HF) communications, satellite navigation and positioning, and radar sensing (Hargreaves, 1992). Therefore, accurately describing the F2-layer electron density profile, particularly the bottomside structure below the peak height, is important for space weather monitoring and the calibration of radio systems. Although ionosonde observations can provide electron density profiles below the peak height of the ionospheric F2 layer, they are inherently in the form of discret. altitude-related measurements. Many practical applications, such as global three-dimensional ionospheric reconstruction and empirical climatological modeling (Radicella et al., 2021), as well as ray-tracing simulations (Tsai et al., 2010), require compact and continuous analytical representations of the ionospheric profile rather than discrete data points alone. To describe the F2-layer bottomside profile in a standardized and continuous manner, the International Reference Ionosphere (IRI) introduces an analytical profile model based on the Epstein function. This model is governed by two key parameters: the thickness parameter *B*0 and the shape parameter *B*1, which characterize the scale of electron density decay and the profile shape below the peak height (Bilitza, 2001). At present, *B*0 and *B*1 are mainly obtained through two approaches.

One approach is empirical-model parameterization, among which the ABT-2009 model is the most representative. Constructed from monthly mean profiles from 27 global ionosonde stations during 1998–2006 using spherical harmonic analysis (Altadill et al., 2009), this climatological model serves as the default option in the current version of IRI (IRI-2020) (Iswariya et al., 2025). However, the ionosphere exhibits pronounced hourly and daily variability, and such climatological-mean models cannot accurately capture real-time disturbances, which can lead to biases when describing instantaneous profiles at specific times and locations.

The second approach relies on real-time observations from ionosondes. As a widely deployed digital ionosonde system, the Digisonde network provides discrete true-height electron density profiles processed by the Automatic Real-Time Ionogram Scaler with True Height (ARTIST) software (Reinisch & Huang, 1983). These

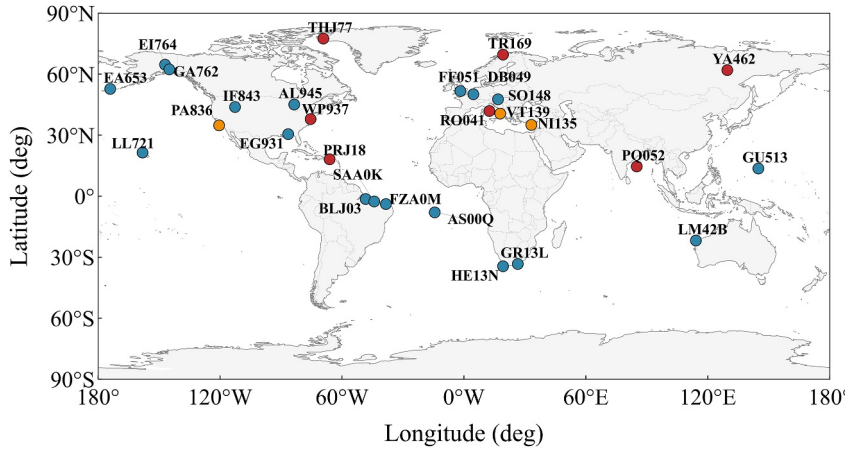


Figure 1. Geographic distribution of the ionosonde stations used in 2014 and 2020.

reconstructed electron density profiles are widely regarded as important observational references for studies of ionospheric vertical structure and bottomside morphology. In addition to the profile data themselves, ARTIST also provides corresponding $B0$ and $B1$ estimates through profile fitting procedures. However, the retrieval of $B0$ and $B1$ in ARTIST is indirect and non-jointly optimized. Following Reinisch and Huang (1998), both the reconstructed true-height profiles and the IRI bottomside profile function are first represented using Chebyshev polynomial expansions, after which least-squares fitting is performed in the polynomial coefficient space. The parameter $B1$ is determined first through iterative searching, while $B0$ is subsequently derived from the fitted coefficients. Such a sequential and polynomial-based parameterization strategy may introduce representation and truncation errors associated with finite-order polynomial approximation, and may limit the ability of the solution to fully capture the global optimum corresponding to the original electron density profile (Huang & Reinisch, 1996).

Given the limited real-time capability of empirical models and the indirect nature of the parameter derivation in ARTIST, this study proposes a framework that uses ionosonde-observed electron density profiles as the reference and directly fits the standard IRI bottomside analytical function for parameter inversion. Methodologically, robust nonlinear least squares (NLS) is first used to obtain physically reasonable and stable initial estimates. The parameters are then refined using four optimization algorithms: gradient descent (GD), Adaptive Gradient Algorithm (AdaGrad), Adaptive Moment Estimation (Adam), and limited-memory BFGS (L-BFGS). Their convergence efficiency and final fitting accuracy are then compared. After the best-performing algorithm is identified, the resulting optimized $B0$ and $B1$ are used as a reference to quantitatively evaluate the parameter biases of the ABT-2009 empirical model. By establishing this high-precision parameter optimization framework, this study can accurately retrieve bottomside ionospheric parameters, thereby providing a reliable data basis for future three-dimensional ionospheric modeling.

2. Data and Methods

2.1. Data Sources

This study employs ionospheric observations from two representative years, 2014 and 2020, corresponding to a period of high solar activity (annual mean sunspot number, SSN ≈ 113.3) and a period of low solar activity (annual mean SSN ≈ 8.8), respectively (SILSO World Data Center, 2025). The analysis is based on vertical electron density profiles of the F2-layer bottomside. The data are from the Global Ionosphere Radio Observatory (GIRO) ionosonde database (Global Ionosphere Radio Observatory, 2026) and cover multiple stations at high, mid, and low latitudes; specifically, 10 stations are used for 2014 and 21 stations for 2020, and the day-of-year (DOY) data corresponding to the March equinox, June solstice, September equinox, and December solstice are extracted for each year (Reinisch & Galkin, 2011). Figure 1 shows the global geographic distribution of the ionosonde stations used in 2014 and 2020: blue circles indicate stations used in 2020, red circles indicate stations used in 2014, and orange circles indicate stations used in both years.

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2.2. Methodology

In this study, the standard analytical function adopted in the IRI model to describe the bottomside profile of the F2 layer is used as the fitting target. This function characterizes the variation of electron density $N_e(h)$ with height h through the thickness parameter $B0$ and the shape parameter $B1$:

$$N_e(h) = \mathrm{NmF2}\,\frac{\exp\left(-X^{B1}\right)}{\cos h(X)} \tag{1}$$

$$X = \frac{\mathrm{hmF2} - h}{B0} \tag{2}$$

Here, NmF2 denotes the F2-layer peak electron density (in cm^{-3}) and hmF2 denotes the corresponding peak height (in km); both are directly obtained from the ionosonde-derived electron density data. The parameter $B0$ is defined as the difference between hmF2 and a characteristic height: in the absence of an F1 layer, this height corresponds to the level where the electron density equals 0.24NmF2. If an F1 layer is present, the F1 peak height is used. $B1$ is a dimensionless shape parameter controlling the curvature of the bottomside profile. Specifically, it describes the morphological characteristics of the electron density profile between the two heights that are used to estimate $B0$ (Altadill et al., 2009).

The estimation of $B0$ and $B1$ can be formulated as a nonlinear least-squares problem. A conventional approach follows the classical iterative linearization idea: a first-order Taylor expansion of the model function is used to linearize the nonlinear model around the current iterate, thereby transforming the original nonlinear problem into a sequence of local linear least-squares problems solved iteratively. However, this approach is sensitive to noise and to data with disparate magnitudes. To obtain a more robust solution, this study first adopts a robust nonlinear least-squares strategy and defines the residual r_i in logarithmic space to minimize relative error:

$$r_i = \log\left(N_{e,i}^{\mathrm{obs}}\right) - \log\left(N_{e,i}^{\mathrm{model}}\right) \tag{3}$$

This formulation shifts the objective toward relative discrepancies, avoids overweighting the high electron density values near the F2 peak in the loss calculation, mitigates imbalance caused by large magnitude differences, and makes the fit more focused on the profile shape itself. Additionally, a soft-L_1 loss function $\rho(r_i)$ is introduced to improve robustness to outliers:

$$\rho(r_i) = 2f_{\mathrm{scale}}^2\left(\sqrt{1 + \left(r_i/f_{\mathrm{scale}}\right)^2} - 1\right) \tag{4}$$

$$w_i = \frac{2}{\sqrt{1 + \left(r_i/f_{\mathrm{scale}}\right)^2}} \tag{5}$$

Here, the weight w_i is given by the derivative of $\rho(r_i)$, which automatically down-weights outliers. Parameter estimation is carried out using an iteratively reweighted Levenberg-Marquardt scheme to progressively optimize an approximately linearized nonlinear model (Marquardt, 1963), yielding initial estimates, denoted $B0_{\mathrm{init}}$ and $B1_{\mathrm{init}}$, that are morphologically closer to the true profile and physically stable and reasonable.

Although this solution is relatively robust, iterative linearization can introduce truncation errors, and a solution under a robust loss does not necessarily correspond to the global optimum in the classical least-squares sense. To further improve inversion accuracy, we apply four modern gradient-based optimization algorithms (GD, AdaGrad, Adam, and L-BFGS), initialized with the NLS solution $B0_{\mathrm{init}}$ and $B1_{\mathrm{init}}$ as a high-quality starting points for parameter refinement. The final fitting objective uses the conventional squared-error loss, in which the observation error is minimized directly without additional weighting:

$$J(B0, B1) = \sum_{i=1}^{n}\left(N_{e,i}^{\mathrm{model}} - N_{e,i}^{\mathrm{obs}}\right)^2 \tag{6}$$

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The selected algorithms differ in their optimization mechanisms. GD, as the most basic first-order method, updates parameters solely based on local gradient information and is sensitive to the choice of learning rate and initial values (Ruder, 2016). AdaGrad and Adam are adaptive learning-rate methods; AdaGrad dynamically adjusts the step size of each parameter by accumulating the squared historical gradients (Duchi et al., 2011). Building on this idea, Adam incorporates estimates of both the first and second moments of the gradients and typically provides better convergence stability for non-stationary objective functions (Kingma & Ba, 2014). L-BFGS is a quasi-Newton method that approximates the inverse Hessian matrix using a limited history of gradients, introducing curvature information without explicitly computing second-order derivatives; it typically achieves faster convergence and higher solution accuracy for medium-scale nonlinear least-squares problems (Nocedal, 1980).

Overall, by initializing with the robust nonlinear least-squares solution and then refining it using these general-purpose gradient-based algorithms, the approach preserves the physical plausibility of the initial estimate while improving the accuracy and stability of the inverted parameters.

2.3. Evaluation Metrics

To quantitatively evaluate the performance of the inversion algorithms and the accuracy of the ABT-2009 model within the IRI framework, we use ionosonde-observed electron density profiles as the reference and compute the mean absolute percentage error (MAPE), mean bias (Bias), and root mean square error (RMSE):

$$\mathrm{MAPE} = \frac{100}{n}\sum_i \left|\frac{N_{e,i}^{\mathrm{model}} - N_{e,i}^{\mathrm{obs}}}{N_{e,i}^{\mathrm{obs}}}\right| \tag{7}$$

$$\mathrm{Bias} = \frac{1}{n}\sum_i \left(N_{e,i}^{\mathrm{model}} - N_{e,i}^{\mathrm{obs}}\right) \tag{8}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i \left(N_{e,i}^{\mathrm{model}} - N_{e,i}^{\mathrm{obs}}\right)^2} \tag{9}$$

When comparing different optimization algorithms, we treat the ionosonde-retrieved electron density profile $N_e^{\mathrm{obs}}(h)$ as the reference, compute the error of the reconstructed profile $N_e^{\mathrm{model}}(h)$, and then take the overall mean across all test profiles. These metrics collectively reflect the accuracy and numerical stability of the algorithms in fitting the vertical structure of the electron density profile. In evaluating the ABT-2009 empirical model, we mainly use Bias and RMSE to quantify the errors of the model outputs.

3. Results and Analysis

3.1. Profile Comparison

To demonstrate the fitting capability of different methods for the F2-layer bottomside profile, electron density profiles reconstructed using seven $B0$ and $B1$ parameter sets are compared with the observed discrete profiles. These seven parameter sets include the native parameters directly output by the ionosonde, the empirical parameters from the ABT-2009 model, the initial parameter solutions obtained via NLS, and the parameters further optimized by GD, Adam, AdaGrad, and L-BFGS algorithms. It should be noted that for the reconstructed profiles of these seven parameter sets, both the peak parameters NmF2 and hmF2 are uniformly fixed to the observed values derived from the ionosonde. This setup ensures that the subsequent profile reconstruction and error assessment focus on reflecting the performance of the bottomside thickness parameter $B0$ and the shape parameter $B1$, isolating the two from the uncertainty of the peak parameters. Figure 2 shows profile comparisons at several representative epochs in years of high and low solar activity, where the black open circles labeled "Observed" represent the discrete true-height electron density data derived from the ionosonde, and Table 1 provides the corresponding RMSE statistics. For the high solar activity year (2014), electron density profiles at station VT139 are selected for the autumn equinox (10:00 UTC) and the winter solstice (17:00 UTC), whereas for the low solar activity year (2020), profiles at station AL945 are selected for the summer solstice and the autumn equinox (both at 10:00 UTC).

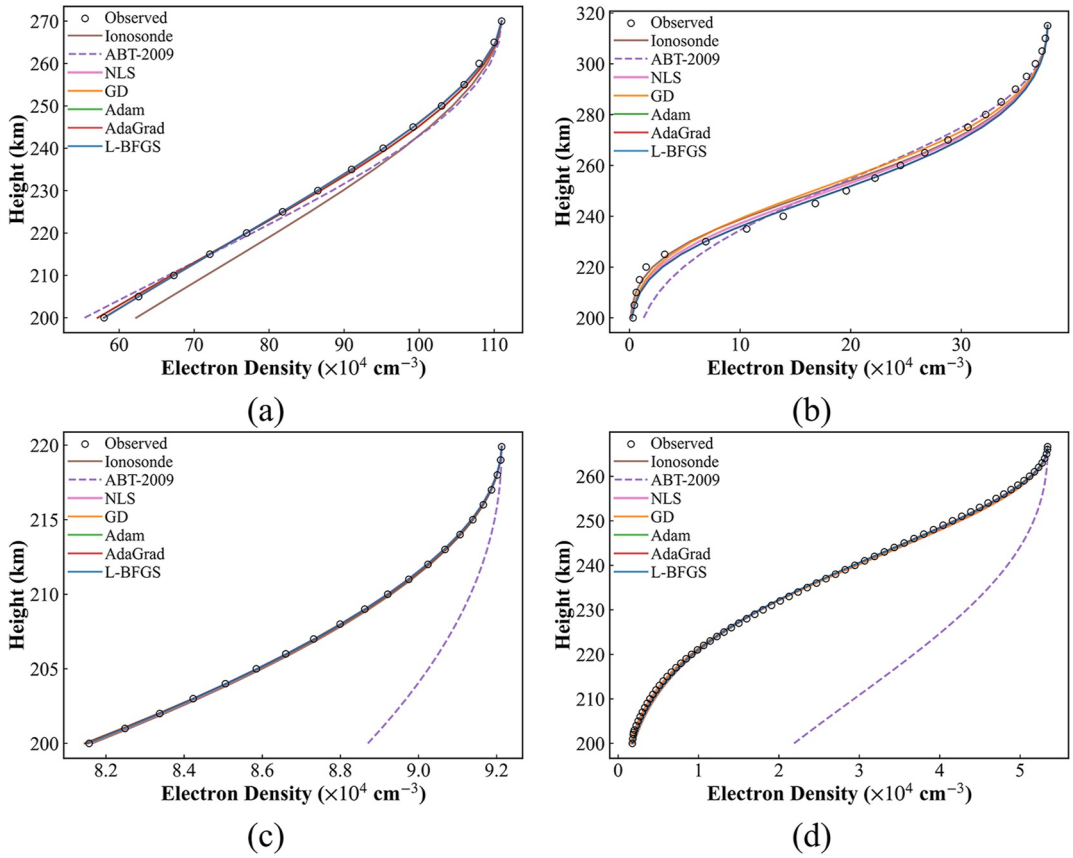


Figure 2. Comparison between reconstructed electron density profiles produced by different algorithms and ionosonde observations under different solar activity levels: (a) VT139 station on the autumn equinox in 2014 (10:00 UTC); (b) VT139 station on the winter solstice in 2014 (17:00 UTC); (c) AL945 station on the summer solstice in 2020 (10:00 UTC); (d) AL945 station on the autumn equinox in 2020 (10:00 UTC).

Considering the combined results from both years, the inversion accuracies achieved by different optimization algorithms are similar in most cases. Except for ABT-2009, which shows clear departures across multiple altitude ranges, the profiles obtained from robust NLS initialization followed by further optimization generally exhibit good morphological agreement with the observations. In most examples, the reconstructed curves from different gradient-based optimizers nearly coincide, indicating that when the initial parameters are already close to the optimum, subsequent optimization tends to converge to the same or neighboring regions of the solution space. The RMSE statistics further reveal differences in stability among the algorithms. L-BFGS yields the lowest errors in all four test cases, showing the most consistent performance. By contrast, Adam and AdaGrad show noticeably

Table 1
RMSE Statistics for Ionospheric Profile Reconstruction Under Different Solar Activity Conditions (Unit: cm^{-3})

	VT139 high solar activity (2014)		AL945 low solar activity (2020)	
Optimizer/Model	Autumn equinox (10:00 UTC)	Winter solstice (17:00 UTC)	Summer solstice (10:00 UTC)	Autumn equinox (10:00 UTC)
Ionosonde	30302.47	12772.62	338.52	91.96
ABT-2009	16744.51	15597.62	20167.10	3501.77
NLS	1311.15	9935.51	412.72	20.08
GD	1311.15	14782.89	413.53	20.08
Adam	4692.58	8909.90	280.96	42.41
AdaGrad	4692.57	8909.90	280.96	35.19
L-BFGS	1310.90	8909.90	280.96	20.04

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Table 2
Performance Comparison of Optimizers Using the 2014 Data Set

Optimizer/Model	MAPE (%)	Bias (cm^{-3})	RMSE (cm^{-3})	Opt. Time (s)
Ionosonde	5.29	−2583.91	14924.25	–
ABT-2009	47.40	−61761.67	92127.73	–
NLS	3.78	593.08	11034.98	–
GD	4.28	1519.16	12189.34	0.88
Adam	2.81	−1291.46	7328.81	2.44
AdaGrad	2.82	−1340.29	7367.59	8.02
L-BFGS	2.64	−886.97	6516.53	0.22

larger errors for some profiles, indicating that adaptive learning-rate methods are more sensitive to nonlinear profile characteristics and initial conditions and can deviate from the optimum in more complex cases. The first-order GD method shows marked degradation for highly nonlinear profiles, which further highlights the need to incorporate curvature information to improve the stability of parameter inversion.

In addition, the RMSE of the NLS-initialized solution is lower than that of ABT-2009 for all four representative profiles and is lower than that of the ionosonde-provided parameters in most cases; the corresponding reconstructed curves also follow the overall trend of the observed profiles. This indicates that the NLS-based result is a reasonable initializer, providing starting parameters closer to the observed profile and thereby improving the overall efficiency and stability of subsequent gradient-based optimization.

3.2. Performance Evaluation

To quantitatively compare the overall performance of different optimization algorithms in electron density profile inversion, we compute the mean MAPE, Bias, and RMSE for each method using all profile samples from 2014 to 2020; these metrics quantify, respectively, the overall relative error of the reconstructed profiles, the systematic deviation, and the error level. Tables 2 and 3 summarize the overall accuracy statistics of the reconstructed electron density profiles for the 2 years.

At the full-sample scale, the gradient-based optimizers show clear differences in the stability of their error statistics under different solar activity conditions. Considering MAPE, RMSE, and Bias jointly, L-BFGS shows higher inversion accuracy and computational efficiency for both years: its MAPE is 2.64% (2014) and 2.35% (2020), its RMSE is 6.5×10^3 and 2.7×10^3 cm^{-3}, and its Bias is lower than that of the other methods; furthermore, L-BFGS has the shortest convergence time among the tested optimizers.

By contrast, the $B0$ and $B1$ parameters prescribed by ABT-2009 yield substantially larger errors, with MAPE generally exceeding 45% and Bias on the order of $10^4 - 10^5$, indicating that climatology-based empirical parameterizations may have limited ability to represent bottomside structural features under actual ionospheric conditions. The parameters directly output by the ionosonde processing software are more accurate than the empirical model, but still show systematic deviations. The statistics also show that the NLS solution used in this

Table 3
Performance Comparison of Optimizers Using the 2020 Data Set

Optimizer/Model	MAPE (%)	Bias (cm^{-3})	RMSE (cm^{-3})	Opt. Time (s)
Ionosonde	5.75	1626.18	7778.69	–
ABT-2009	57.26	−24595.88	44572.16	–
NLS	3.53	1328.39	4878.88	–
GD	3.91	1551.32	5192.28	3.61
Adam	2.76	−702.18	4236.84	7.48
AdaGrad	2.73	−678.71	4140.51	22.33
L-BFGS	2.35	224.13	2740.34	0.61

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study improves the accuracy of the initial parameter estimates and provides a stable and physically reasonable starting point for subsequent gradient-based refinement.

Overall, L-BFGS shows the best overall performance for ionospheric morphological parameter inversion, and its advantages in accuracy, stability, and computational efficiency make it suitable as the reference solution for the quantitative evaluation of systematic biases in ABT-2009 in the subsequent sections.

3.3. Evaluation of the ABT-2009 Model

In this section, the L-BFGS-optimized results from Section 3.2 are used as the benchmark to evaluate the applicability of the ABT-2009 model for the morphological parameters $B0$ and $B1$ under different geographic locations and seasonal conditions worldwide. The evaluation uses Bias and RMSE. For the i-th sample, the deviation ΔP for a given parameter P is defined as

$$\Delta P = P^{\mathrm{ABT}} - P^{\mathrm{L-BFGS}} \tag{10}$$

where P represents the ionospheric structural parameters $B0$ and $B1$. Here, P^{ABT} denotes the prediction from the ABT-2009 model, and $P^{\mathrm{L-BFGS}}$ denotes the inverted result which is treated as the benchmark. Unlike the profile reconstruction metrics in Section 3.2, the Bias and RMSE reported in this section are computed directly for the retrieved $B0$ and $B1$ parameters. Bias characterizes the systematic offset of the model, whereas RMSE quantifies the dispersion of errors and the overall accuracy. It should be noted that in the empirical formulation of the ABT-2009 model, $B0$ and $B1$ are generated independently via spherical harmonic functions driven by coordinates, time, and solar activity, without utilizing hmF2 or NmF2 as inputs (Altadill et al., 2009). Therefore, the parameter errors quantified in this section are inherently attributable to the ABT-2009 model itself and are not directly associated with the prediction biases of the F2-layer peak parameters.

3.3.1. Latitudinal Bands and Seasonal Statistic

Samples are grouped into three latitude bands: low latitudes (<30°), mid-latitudes (30°–60°), and high latitudes (>60°). Profiles are further classified into four seasonal groups corresponding to the two equinoxes and two solstices. Figure 3 shows the statistical distributions of $B0$ and $B1$ errors across latitude bands and seasons under different solar activity levels. It should be noted that since $B0$ is a thickness parameter, its associated Bias and RMSE are evaluated in kilometers (km). Conversely, as the shape parameter $B1$ is inherently dimensionless in the standard IRI formulation, its corresponding statistical metrics presented in the figures do not carry physical units.

In 2014, the model shows an overall predominance of positive Bias in $B0$. Errors at mid-latitudes are relatively low and show good stability. By contrast, errors are larger at low and high latitudes; at high latitudes in summer, the $B0$ Bias changes sign from overestimation to underestimation and the dispersion increases. For $B1$, the model performs better at low latitudes and shows a slight positive Bias at mid-latitudes, while the overall errors remain relatively stable. At high latitudes, $B1$ exhibits a marked negative Bias; in winter, abnormally large error values occur. The larger errors of $B0$ and $B1$ at high latitudes may be related to polar ionospheric conditions influenced by magnetospheric particle precipitation, where profile gradients are steep and temporal variability is large, which is difficult for empirical models to represent accurately. Seasonally, the model is more stable for both parameters around the autumn equinox, whereas the spring equinox and winter solstice are often associated with larger errors. Specifically, $B0$ shows very large positive Bias at the spring equinox at both high and low latitudes. The winter solstice represents a failure point for high-latitude $B1$, producing extreme error peaks.

As shown in Figure 3, during the low solar activity period, ABT-2009 retains an overall tendency to overestimate $B0$, while the systematic Bias improves at low latitudes and the overall mean overestimation decreases to 6.58 km. For $B1$, although the overall Bias is small, accuracy decreases, and larger RMSE values occur at mid-to-low latitudes. Seasonally, $B0$ remains overestimated at high latitudes at the spring equinox and winter solstice, while mid-to-low latitudes show small Bias but large RMSE at the summer solstice. $B1$ does not show extreme seasonal fluctuations. Bias remains close to zero throughout the year, indicating very weak systematic deviation, yet RMSE persistently stays at a relatively high level (approximately 1.3–1.8) without a clear seasonal preference; moreover, accuracy at mid-to-low latitudes is even lower than that under high solar activity. This pattern of small Bias but large RMSE is consistent for $B0$ in low-latitude summer and for low-latitude $B1$ across all seasons. This

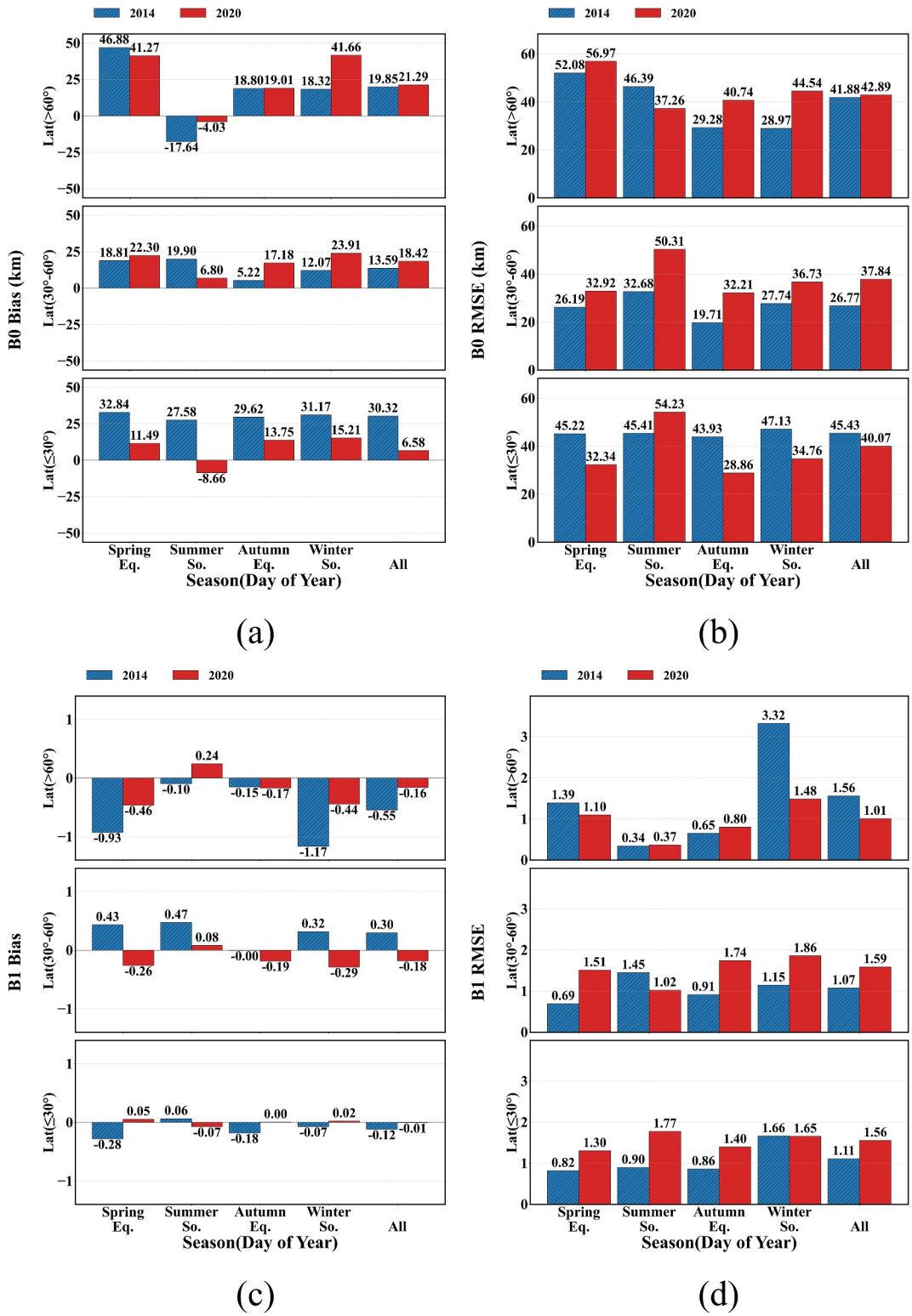


Figure 3. Statistical distributions of ABT-2009 model errors for parameters *B*0 and *B*1 across seasons and latitude bands under different solar activity conditions: (a) overall mean Bias of *B*0; (b) overall root mean square error (RMSE) of *B*0; (c) overall mean Bias of *B*1; (d) overall RMSE of *B*1.

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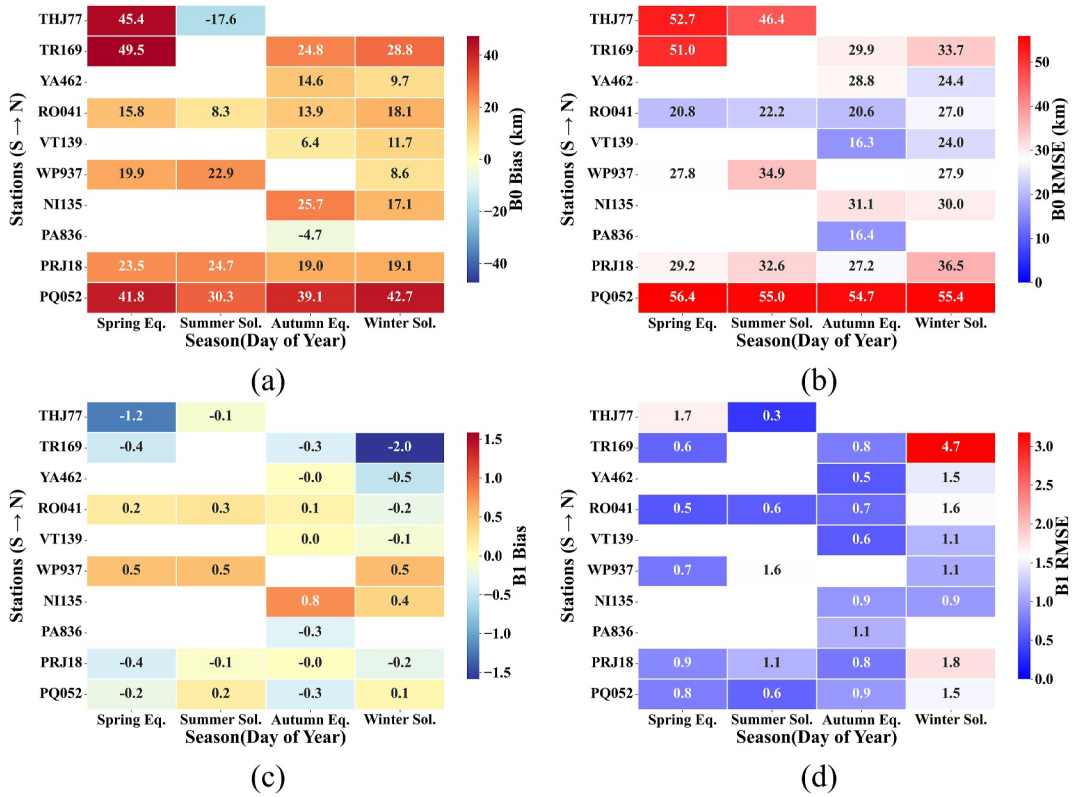


Figure 4. Spatiotemporal heatmaps of ABT-2009 model prediction errors for 2014 (high solar activity year): (a) mean Bias of $B0$; (b) root mean square error (RMSE) of $B0$; (c) mean Bias of $B1$; (d) RMSE of $B1$. Note: Stations on the y-axis are ordered by geographic latitude from south to north, and the x-axis denotes season.

behavior reflects a mismatch between the model formulation and the observation time scale: ABT-2009 is constructed as a climatological background mean of the ionosphere and can therefore agree with the reference values in a statistical-mean sense, but it may not capture finer temporal variations of instantaneous bottomside profiles, leading to large scatter between predictions and observations.

3.3.2. Station-Based Error Statistics

To further examine the spatial distribution of model errors, Figures 4 and 5 present heatmaps of station-wise errors in $B0$ and $B1$ for the high and low solar activity years, respectively. It should be noted that the station sets in 2014 and 2020 are not identical; therefore, the comparison focuses on regional-scale error structures rather than year-to-year changes at individual stations. In the figures, stations are ordered by latitude from low to high (i.e., from south to north). All data used for 2014 are from Northern Hemisphere stations, and blank cells indicate missing data for the corresponding periods.

Low latitudes show the most concentrated errors in $B0$, but the error patterns differ between solar activity conditions. During the high solar activity year (2014), the representative station PQ052 shows a pronounced positive Bias and relatively large RMSE throughout the year. During the low solar activity year (2020), both $B0$ and $B1$ show more prominent seasonal features. At stations such as GU513, SAA0K, and FZA0M, RMSE increases markedly around the local summer solstice. Moreover, Bias shows clear hemispheric asymmetry: mid-to-low latitudes in the Northern Hemisphere tend to be overestimated in summer, whereas low latitudes in the Southern Hemisphere show systematic underestimation. This hemispheric contrast suggests that, under low solar activity, ABT-2009 may be limited in its parameterization of interhemispheric plasma transport and the seasonal morphology of the low-latitude ionosphere.

Mid-latitude stations generally have smaller biases and are a relatively robust region for the model, but more evident seasonal errors appear in the low solar activity year. In 2014, most mid-latitude stations (e.g., RO041 and

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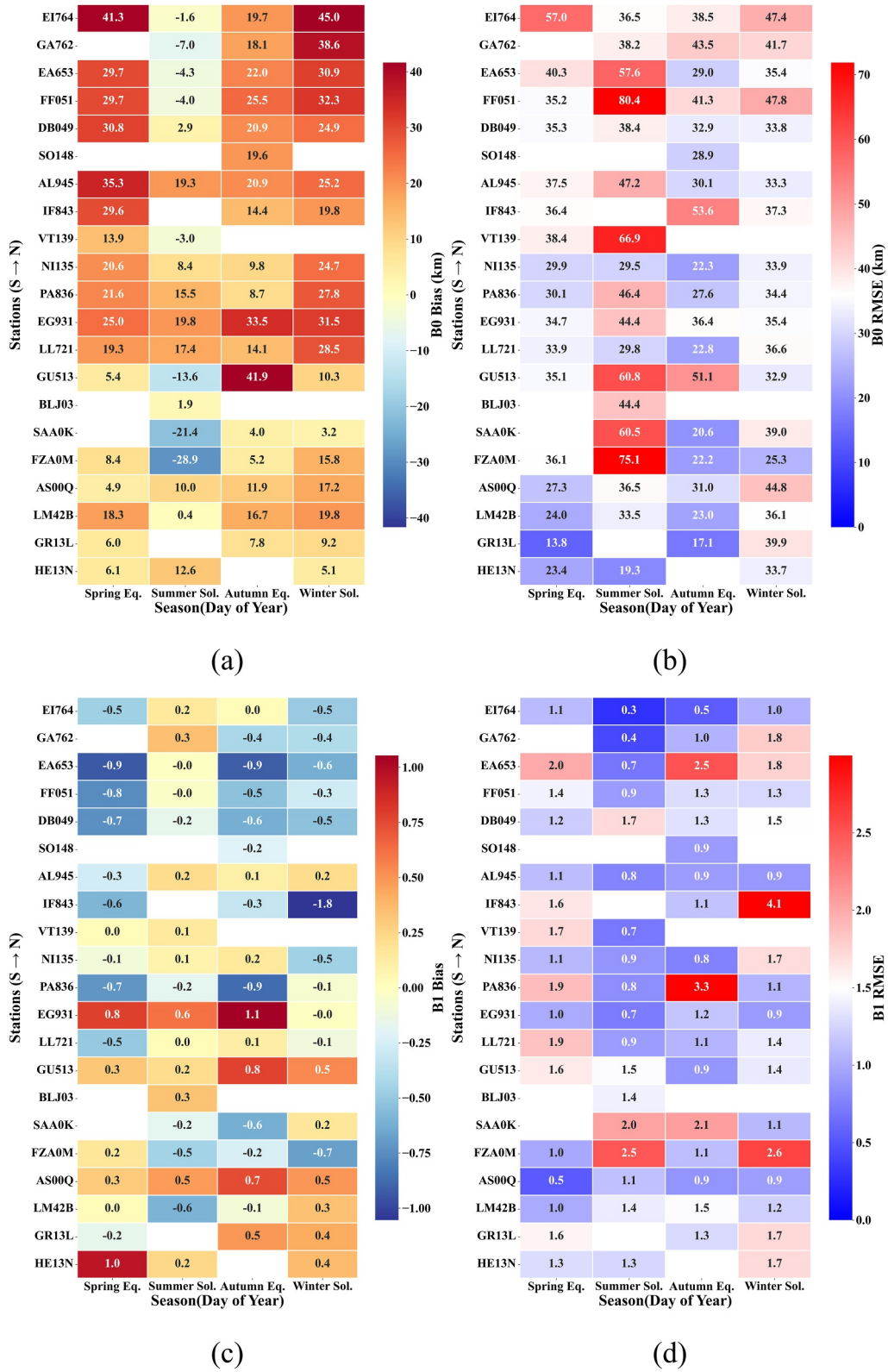


Figure 5. Spatiotemporal heatmaps of ABT-2009 model prediction errors for 2020 (low solar activity year). Panels (a)–(d) have the same definitions as in Figure 4.

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VT139) maintain low Bias and RMSE for both parameters, indicating the most stable model performance. In 2020, by contrast, several mid-latitude stations (e.g., EA653, FF051, and VT139) show markedly higher RMSE during local summer, with a particularly pronounced increase in the dispersion of $B0$. The Bias of $B1$ remains close to zero throughout the year at most mid-latitude stations. A localized winter RMSE peak is observed at the North American mid-latitude station IF843.

High-latitude stations exhibit strong seasonal error structures. Consistent with the statistics in Section 3.3.1, error maxima for both $B0$ and $B1$ are concentrated around the spring equinox and winter solstice, although the two parameters respond differently to solar activity levels. Stations THJ77 and TR169 in 2014, as well as station EI764 in 2020, show pronounced positive Bias accompanied by increased RMSE at the spring equinox and winter solstice. Unlike the persistent Bias in $B0$, $B1$ shows an extreme RMSE value at station TR169 during the winter solstice in 2014, whereas the corresponding $B1$ extreme in 2020 is clearly reduced.

4. Conclusions

This study investigated the inversion of the ionospheric F2-layer bottomside morphological parameters $B0$ and $B1$ and developed an inversion framework that used NLS to obtain initial estimates and then refined the parameters using gradient-based optimization algorithms. The results showed that NLS provides physically reasonable and stable initial parameters under a range of ionospheric conditions, offering a reliable starting point for subsequent optimization, and that gradient-based refinement can further reduce inversion errors. A comparison among the optimization algorithms indicated that L-BFGS offers a more balanced performance in terms of accuracy, convergence efficiency, and stability; therefore, the NLS-initialized L-BFGS strategy was used as the basis for subsequent model evaluation.

Using this inversion strategy, this study further examined the performance of the ABT-2009 model for $B0$ and $B1$ under different ionospheric conditions. The results indicated that in the high solar activity year, $B0$ is dominated by positive Bias at the global scale, with relatively stable behavior at mid-latitudes but larger errors at low and high latitudes, and a summer transition at high latitudes from overestimation to underestimation; $B1$ is generally stable at low and mid-latitudes, whereas negative Bias and error peaks occur more readily at high latitudes, suggesting limited applicability in polar regions. In the low solar activity year, $B0$ errors increase in summer at mid-latitudes and localized anomalies occur more frequently, indicating that the model is less responsive to short-term profile variability under relatively quiet conditions; meanwhile, $B1$ more typically exhibits the pattern of "Bias near zero but persistently high RMSE," pointing to a discrepancy between agreement in the mean and the reproduction of instantaneous profile structures. The heatmaps further indicate that errors remain concentrated throughout the year at low latitudes, that mid-latitudes are more strongly modulated by seasonal forcing under low solar activity, and that high latitudes consistently exhibit Bias peaks around the spring equinox and winter solstice. Overall, the applicability of the ABT-2009 model differs across latitude bands and seasons.

In summary, based on discrete true-height ionosonde observations, this study systematically analyzed parameterization methods for the F2-layer bottomside profile. The results demonstrated that the proposed parameter optimization framework can successfully retrieve bottomside ionospheric morphological parameters with high precision. As global ionosonde observation networks continue to expand and accumulate more data in the future, this methodology will enable the extraction of more abundant and precise $B0$ and $B1$ parameters on a global scale, thereby laying a solid and scalable data foundation for the development of next-generation three-dimensional ionospheric models. Furthermore, considering the physical correlation between the bottomside shape parameters and the peak parameters, our future research will expand upon this framework. By establishing and integrating models for both the bottomside ($B0$, $B1$) and peak parameters (NmF2, hmF2), we plan to systematically evaluate how their combined errors jointly impact the overall representation of the bottomside electron density profile.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

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Availability Statement

The ionosonde observation data used in this study were downloaded from the Digital Ionogram DataBase (DIDBase) of the Global Ionosphere Radio Observatory (GIRO) via the official SAO Explorer software (Global Ionosphere Radio Observatory, 2026). The downloaded data, provided in SAO and SAO.XML formats, were parsed to extract the required electron density profiles and F2-layer morphological parameters. The specific observation stations and time periods are detailed in Section 2.1 and Figure 1 of this manuscript. The annual mean sunspot number data utilized to define the solar activity levels for 2014 and 2020 are openly available from the SILSO World Data Center (SILSO World Data Center, 2025).

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